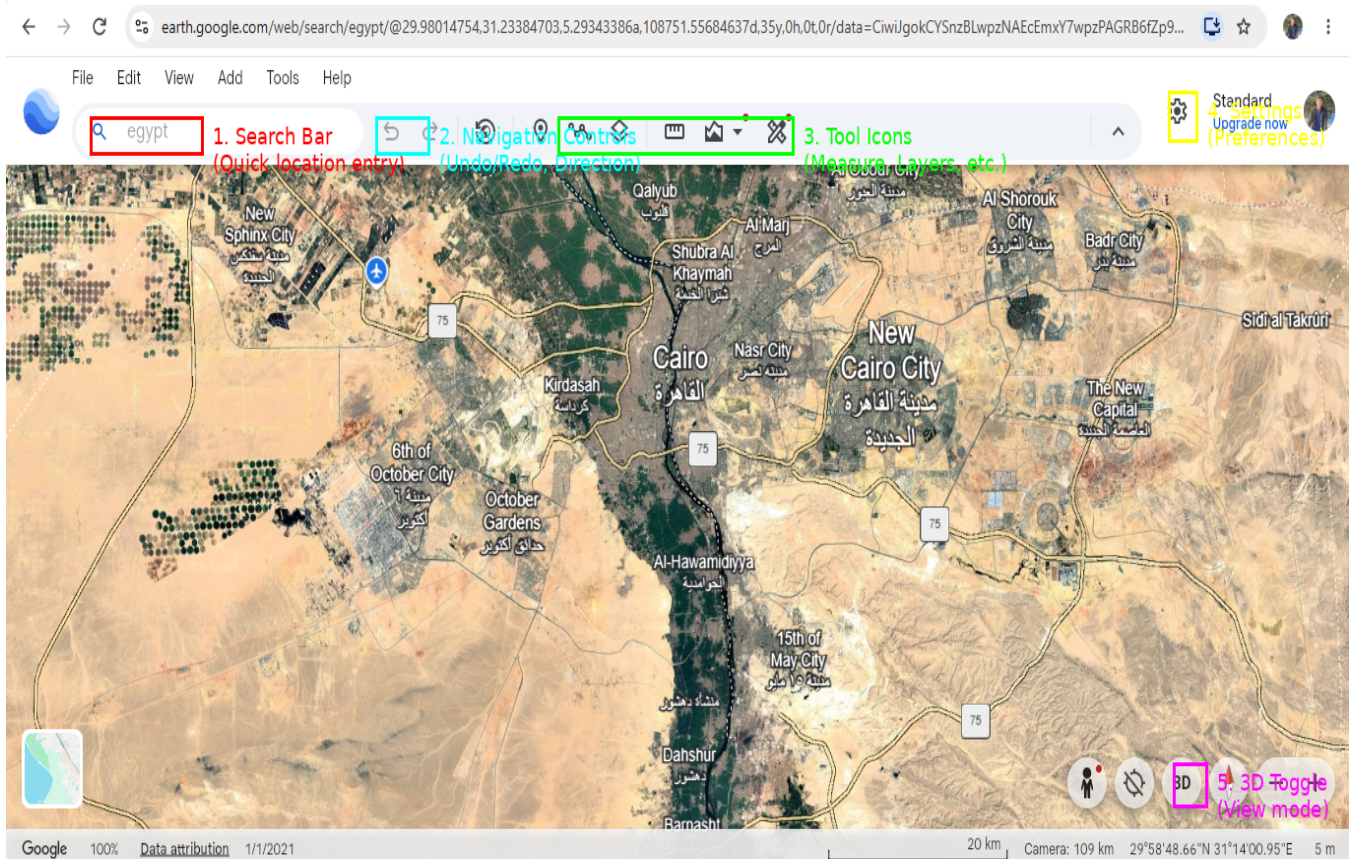


Google Earth: HCI vs Computer Graphics Analysis

SLIDE 1: HCI (Human-Computer Interaction) Analysis

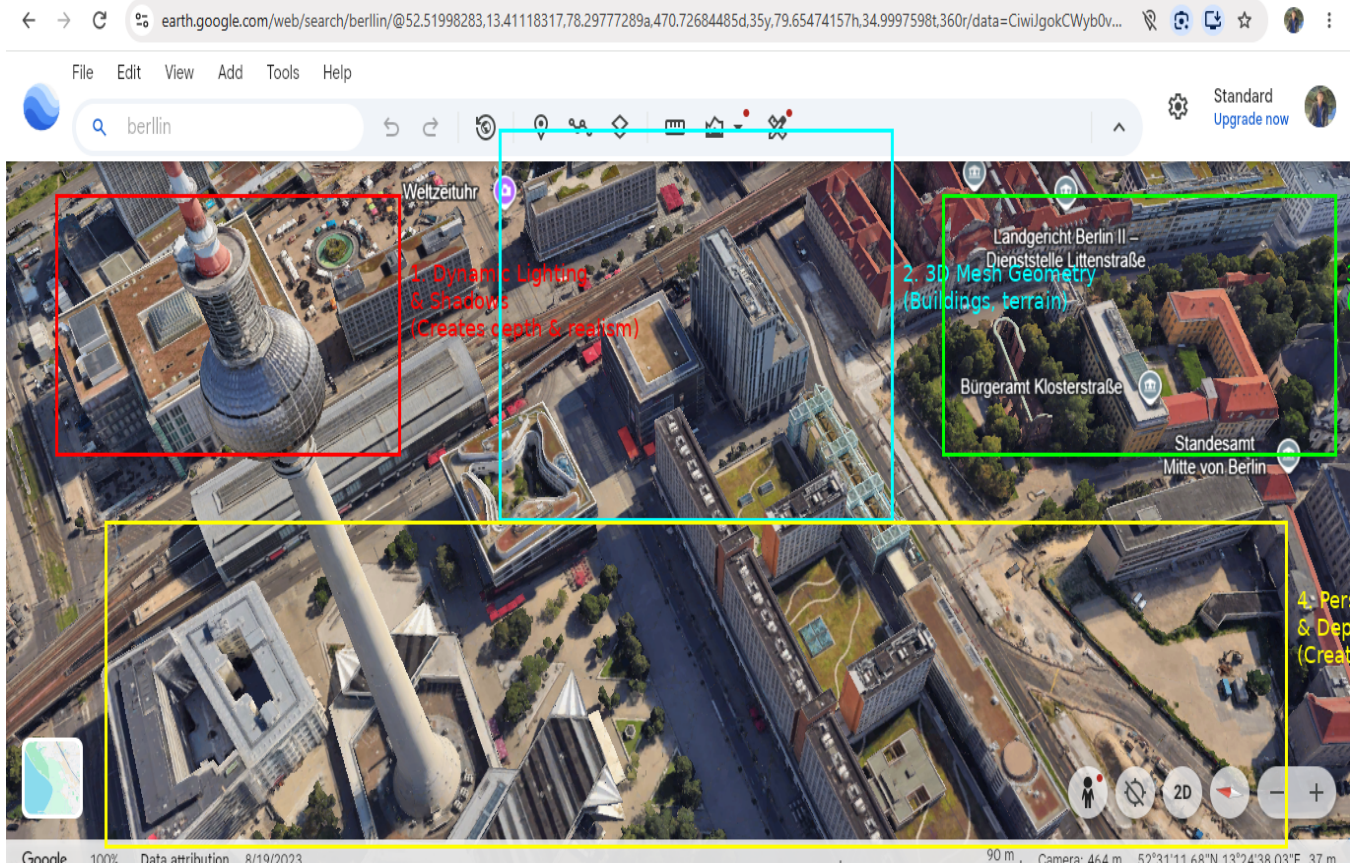


HCI Elements Identified:

- 1. Search Bar (Red)** - Allows users to quickly navigate to any location on Earth by typing place names, coordinates, or addresses. Provides immediate feedback and is the primary way users explore specific areas.
- 2. Navigation Controls (Cyan)** - Undo/redo buttons and directional arrows help users navigate smoothly through their exploration. These controls provide quick access to previous views without having to re-search.
- 3. Tool Icons (Green)** - Measure distance, draw shapes, view layers, and access Street View. These tools give users clear visual affordances about what actions are possible, improving usability.
- 4. Settings Icon (Yellow)** - Provides access to display options and preferences. Clear iconography helps users understand this controls the app's behavior and appearance.
- 5. 3D Toggle Button (Magenta)** - Switches between 2D and 3D view modes. This is critical HCI

because it lets users choose their preferred visualization style based on their task.

SLIDE 2: Computer Graphics (CG) Analysis



Graphics Rendering Elements:

1. Dynamic Lighting & Shadows (Red) - Real-time shadows cast by buildings create depth perception and make the 3D space feel realistic. The sun position affects shadow direction, adding temporal realism to the visualization.

2. 3D Mesh Geometry (Cyan) - Buildings and terrain are represented as 3D polygon meshes. Google Earth uses satellite imagery and LiDAR data to construct accurate 3D models of cities, making the environment believable.

3. Texture Mapping (Green) - High-resolution satellite and aerial photographs are applied as textures onto the 3D geometry. This creates photorealistic surfaces while keeping file sizes manageable through GPU compression.

4. Perspective Camera & Depth Cues (Yellow) - The 3D perspective projection creates visual depth through: relative size (farther objects appear smaller), occlusion (buildings hide what's behind them), and atmospheric perspective (distant objects are hazier). These cues are essential for perceiving 3D space on a 2D screen.

REFLECTION QUESTIONS & ANSWERS

Q1: How does the Computer Graphics quality (realistic lighting, 3D view) help or hinder the user's ability to interact with the software?

Answer: The high-quality graphics are extremely helpful for interaction. Realistic lighting and shadows provide crucial visual feedback—users can immediately see the terrain elevation and building heights. The 3D perspective helps users understand spatial relationships and navigate more intuitively compared to flat 2D maps. However, in some cases, excessive detail (like texture complexity) could slightly slow down interaction on older devices, but the benefits far outweigh this minor drawback. The photorealistic rendering makes the tool engaging and helps users connect what they see in Google Earth to the real world they know.

Q2: What is one simple HCI change you would make to improve the user interface?

Answer: I would add contextual tooltips that appear when hovering over tool icons. Currently, many users might not know what each icon does without clicking it. Adding small hover labels like "Measure distance," "View layers," and "Street View" would reduce the learning curve and make the interface more accessible to new users. This simple addition would improve discoverability without cluttering the interface, following the principle of progressive disclosure in good HCI design.